

MATH 20C: CALCULUS AND ANALYTIC GEOMETRY FOR SCIENCE AND ENGINEERING

Summer Session II 2026

Instructor:	Casey Perdue	Lecture Time:	MWF 12–1:50pm
Email:	caseyperdue@ucsd.edu	Lecture Place:	HSS 1315
TA:	Lixing Yi	Section Time:	TuTh 1-1:50pm
Email:	liyi@ucsd.edu	Section Place:	HSS 1128A

Note: Please send us emails for urgent questions! We see these much more quickly than Canvas or Piazza private messages.

Office Hours: Casey's OHs: MWF 3-4pm in HSS 5012, or by appointment.
Lixing's OHs: Tuesday 2-3:30 in HSS 3062.

Course Description: Vector geometry, vector functions and their derivatives. Partial differentiation. Maxima and minima. Double integration.

Textbook: Calculus Early Transcendentals: Multivariable with Achieve, Rogawski, 4th edition.
Note: The Math department requires usage of Achieve for 20C homework assignments; you unfortunately must pay to access these assignments.

Time Commitment: Courses during a 5-week summer session move twice as quickly compared to courses during a typical quarter; you should consider this course to be equivalent to an 8 unit course during the academic year in terms of your time commitment. That is, you should expect to spend about 24 hours per week on this course, according to the policy of UCSD's Academic Senate: "One unit for a lecture-based course taught in the summer quarter over the duration of five weeks will consist of two hours of lecture and four hours of out-of-class student preparation per week."

Exams: There will be two in-person exams, one midterm and one final, as follows:

- The midterm is scheduled for **Wednesday, August 19 at 12-1:30pm in HSS 1128A.**
- The final is scheduled for **Saturday, September 5 at 3-6pm in HSS 2150.**

If you cannot attend any of the exams at the scheduled times, please do not take this class; there will be no make-up exams unless any true last-minute emergencies arise (in which case, you must email me ASAP, and we can discuss your situation and determine if a make-up exam is appropriate). During each exam, you **may not bring calculators**, however **you are permitted one 8.5x11in sheet of handwritten notes** (both sides). No other outside materials are allowed. **Please bring your student IDs to the exam, as we will be checking them.**

The midterm will be on all material from the sections we used in Chapters 12 and 13 (12.1, 12.2, 12.3, 12.4, 12.5, 13.1, 13.2, 13.3, 13.5). The final will be cumulative, and will be on all material from the sections we used in Chapters 12, 13, 14, and 15 (the above sections, as well as 14.1, 14.2, 14.3, 14.4, 14.5, 14.6, 14.7, 14.8, 15.1, 15.2).

Homework: Online homework will be done through Achieve. There will be a homework assignment for each of the 19 textbook sections covered. The homework assignments will be due every Friday and Monday, with the exception of the Monday of Week 1, and will become available one week before their due date. You have unlimited attempts for each question, with no penalty. **Note:** The homework assignments will allow

late submissions up to 48 hours past their original due date, with no penalty. (The only exception are the last homework assignments due Friday September 4th, as this grace period would put it due after the final exam. These will not allow any late submissions.) Please be aware that no further accommodations will be granted, beyond the drop policy listed below.

Quizzes: There will be three 20-minute quizzes at the start of lecture (12:00pm-12:20pm) on the following dates:

- Monday, August 10
- Monday, August 24
- Monday, August 31

These are not meant to be intimidating, but are meant to simply supplement the homework and to prepare you for the exams. If you are keeping up with the lectures and are comfortable with the homework, then these should be doable. They will be based on the material covered up to the Wednesday of the prior week. As with the exams, if you cannot attend at least 2 of these quizzes, please do not take this class.

Discussion Attendance: **Attending discussion sections is required**, and will constitute a small portion of your grade. There are 10 discussion sections; we will drop your lowest 1 of 10 attendance scores. That is, if you attend at least 9 sections, you will receive full points. **Attending lecture is not required, however it is strongly encouraged** as the **lectures will not be recorded**. Lecture notes will be posted on Canvas, but this will not be as effective at teaching the material as the actual lecture.

Drop Policy: We will drop the lowest 3 of the 19 homework assignments, the lowest quiz score, and the lowest 1 of 10 attendance scores. The purpose for this drop policy is that we are aware that sometimes circumstances beyond the control of a student will prevent them from completing the work within the allowed time or at the prescribed location.

Grading Scheme: Your final score in this course will be calculated using the higher value of one of the following two schemes:

- 9% Homework + 1% Attendance + 15% Quizzes + 30% Midterm + 45% Final exam
- 9% Homework + 1% Attendance + 15% Quizzes + 10% Midterm + 65% Final exam

Letter grades will be assigned according to the following grading scale:

A+	A	A-	B+	B	B-	C+	C	C-	D	F
[98,100]	[93,98)	[90,93)	[87,90)	[83,87)	[80,83)	[77,80)	[73,77)	[70,73)	[60,70)	[0,60)

There may be a curve applied at the end of the course if any of the assignments were harder than intended, however you should not expect a curve by default.

Note: We will not take into account graduation requirements, personal requests, or the desire for a certain grade for any reason when assigning letter grades. Once assigned, letter grades are final.

AI/Collaboration/Academic Integrity Policy: In this course, **the use of Artificial Intelligence tools such as ChatGPT, Claude, etc is not allowed** when completing the homework. You may use it to assist you while studying, however it is strongly discouraged; please use other resources first, such as course staff's office hours.

You may discuss homework problems with your classmates, however you should not copy answers in any way. Everything that you submit should be your own work and reflect your own understanding of the problems.

According to the UCSD Policy on Integrity of Scholarship, “no student shall engage in an activity that undermines academic integrity or facilitates academic integrity violations by others.” Activities that are explicitly banned include:

- Complete any part of any assignment for someone else.
- Have any part of any of your assignments completed by someone else.
- Plagiarize or copy any part of any of someone else’s work and submit it as your own.
- Employ aids excluded by the instructor in completing any assignment.

Announcements: The announcements page on Canvas will be the regular means of communication with the class. It is your responsibility to check this regularly and you may want to adjust your Canvas notification settings in order to receive notifications that you may be missing.

Regrade Requests: Regrade requests will be made using the built-in regrade request feature in Gradescope. There will be a limited window of time after your graded work is made available during which the regrade request feature will be active. This time window will be announced when the scores are released to the students. Please be advised that the regrade request window is usually brief, and if you want to request a regrade, it is your responsibility to make the request during the allowed time. Please understand that while we will correct errors in the grading, we will not modify the grading rubric or negotiate over partial credit after graded papers are returned to students.

Piazza: The instructor and the TA will periodically check the Piazza page on weekdays. Only comments and questions pertaining to the mathematical content and logistics of the course are allowed; any postings that do not meet this criterion will be deleted. As a diverse community of learners, students must strive to work together in a setting of civility, tolerance, and respect for each other and for the instructors! Conflicting opinions among members of a class are to be respected and responded to in a professional manner.

Accommodations: Students requesting accommodations for this course due to a disability must provide a current Authorization for Accommodation (AFA) letter (paper or electronic) issued by the Office for Students with Disabilities. Students are required to discuss accommodation arrangements with instructors and OSD liaisons in the department **in advance** of any exams or assignments.

Filing for an Incomplete: Sometimes it is not possible to complete all of the work in the course due to circumstances beyond your control. (Being unable to take the final exam because of accident or sickness, for example.) In such a case, it may be possible to file for an Incomplete grade. If an Incomplete is granted, you will be given the opportunity to complete the work during the next quarter. (All work must be completed before the end of Week 10 the following academic quarter.) In order to be eligible for an Incomplete, you must meet the following criteria:

- Your incomplete work must be due to circumstances beyond your control. (Sickness, accident, etc.)
- You must be able to verify the circumstances with official documentation.
- Your course work must be of “non-failing quality” (from the Academic Senate Regulations). This means that you must be currently passing the course according to the method for computing grades listed above.

To apply for an Incomplete, you must fill out and submit the official form.

University-Wide Resources:

- **Basic Needs** at UCSD: Any student who has difficulty accessing sufficient food to eat every day, or who lacks a safe and stable place to live is encouraged to contact basicneeds@ucsd.edu

- **Counseling and Psychological Services (CAPS):** Confidential counseling and consultations for psychiatric service and mental health programming can be found at caps.ucsd.edu
- **Office for Students with Disabilities (OSD):** Supports students with disabilities and accessibility across campus. Can be found at osd.ucsd.edu
- **Undocumented Student Services (USS):** Programs and services designed to help students overcome obstacles that arise from their immigration status. Can be found at uss.ucsd.edu

Course Calendar: Note: The list of topics included below is approximate, and could potentially change during the quarter depending on how quickly we move through them.

Week	List of topics	M	W	F	Sat
1 (8/2-8/8)	12.1 – 12.5				
2 (8/9-8/15)	13.1 – 13.3, 13.5	Quiz at 12pm			
3 (8/16-8/22)	14.1 – 14.4		Midterm at 12pm		
4 (8/23-8/29)	14.5 – 14.8	Quiz at 12pm			
5 (8/30-9/5)	15.1, 15.2	Quiz at 12pm		Review session	Final at 3pm

This syllabus is subject to change during the quarter due to unforeseen circumstances.